VMamba: Visual State Space Model

NeurIPS 2024 / 1465 citations (~4/28)

Hyunoh Kang

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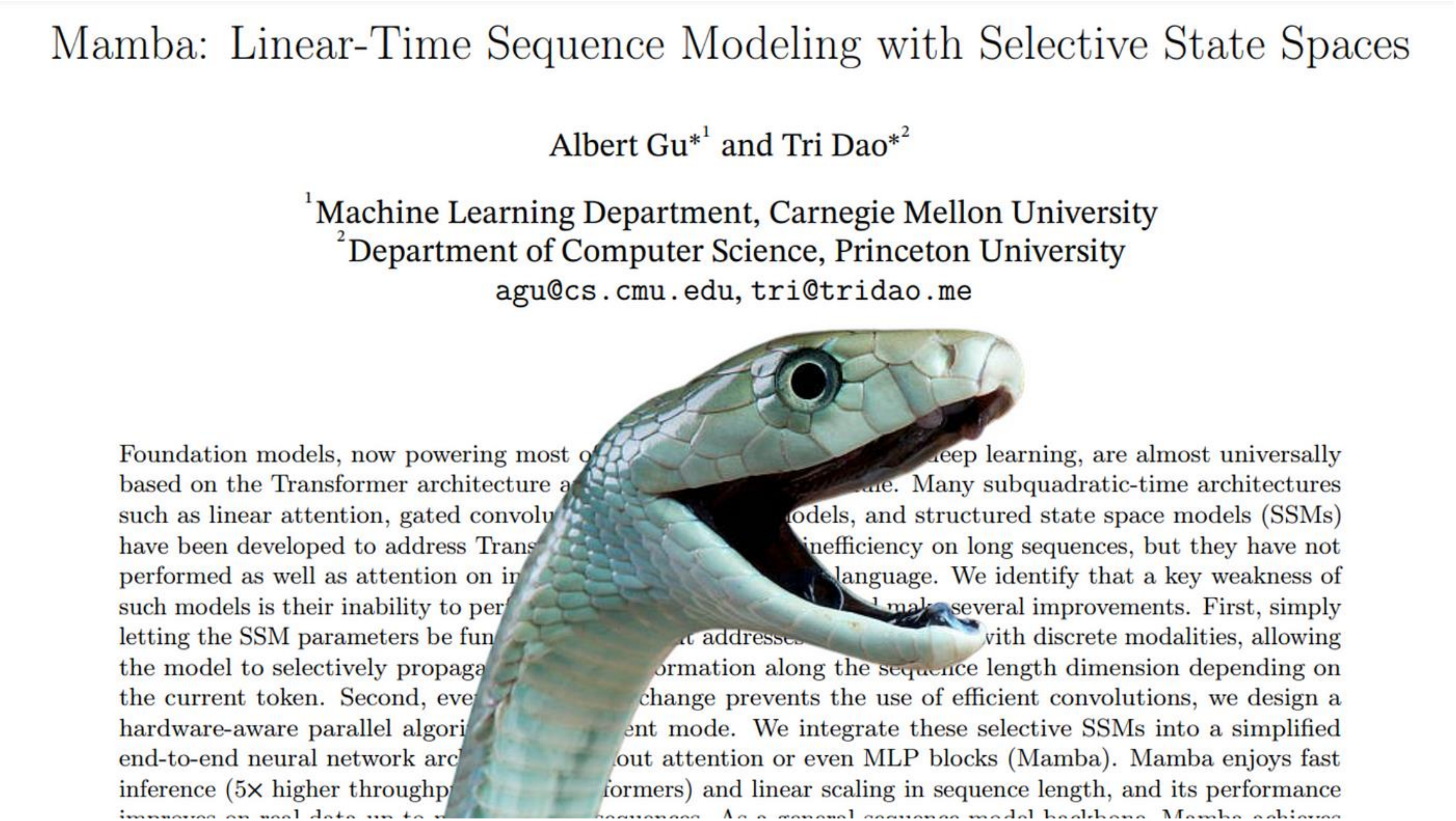
Introduction



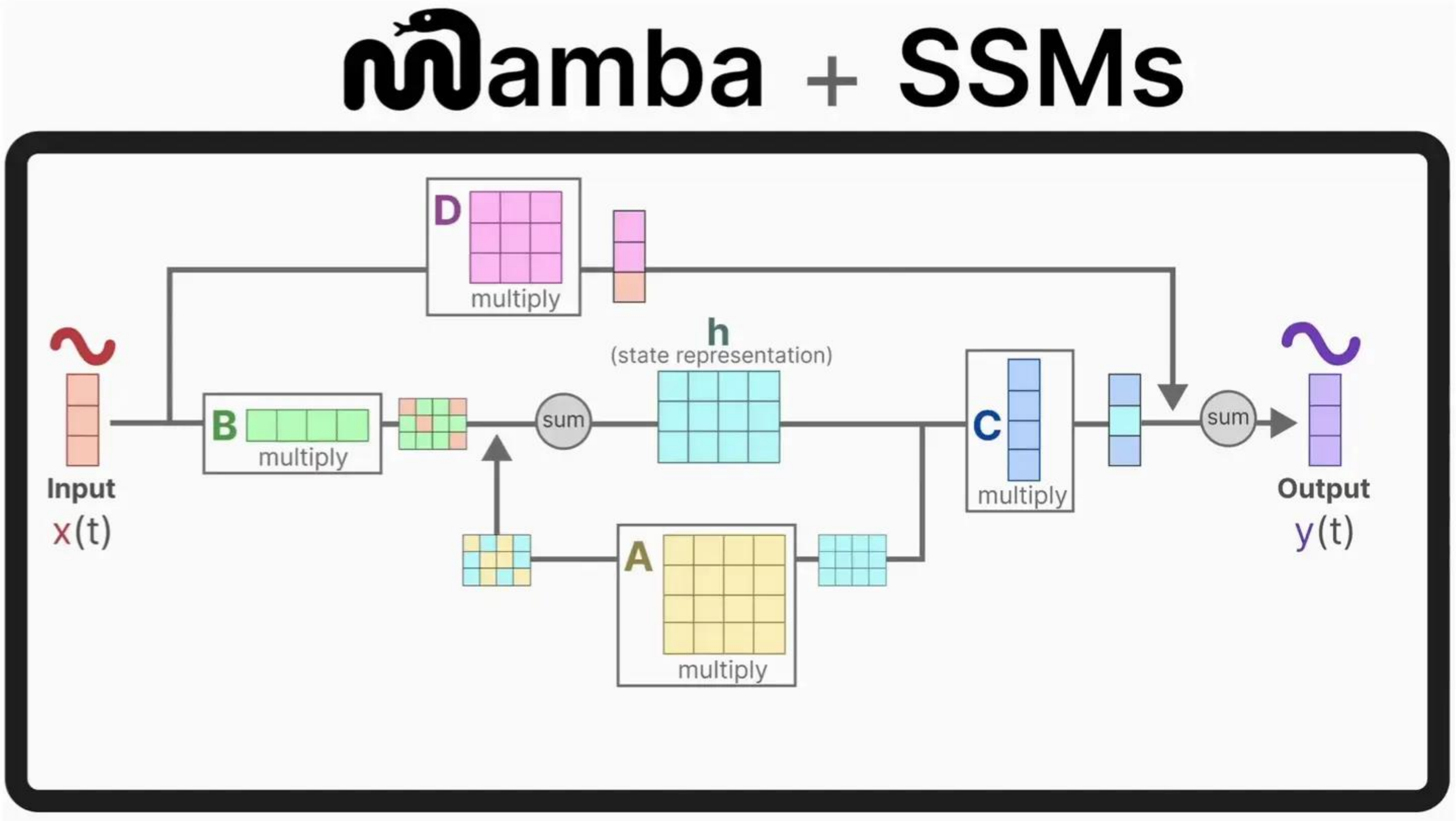
Mamba: Linear-Time Sequence Modeling with Selective State Spaces

Albert Gu*[1] and Tri Dao*[2]

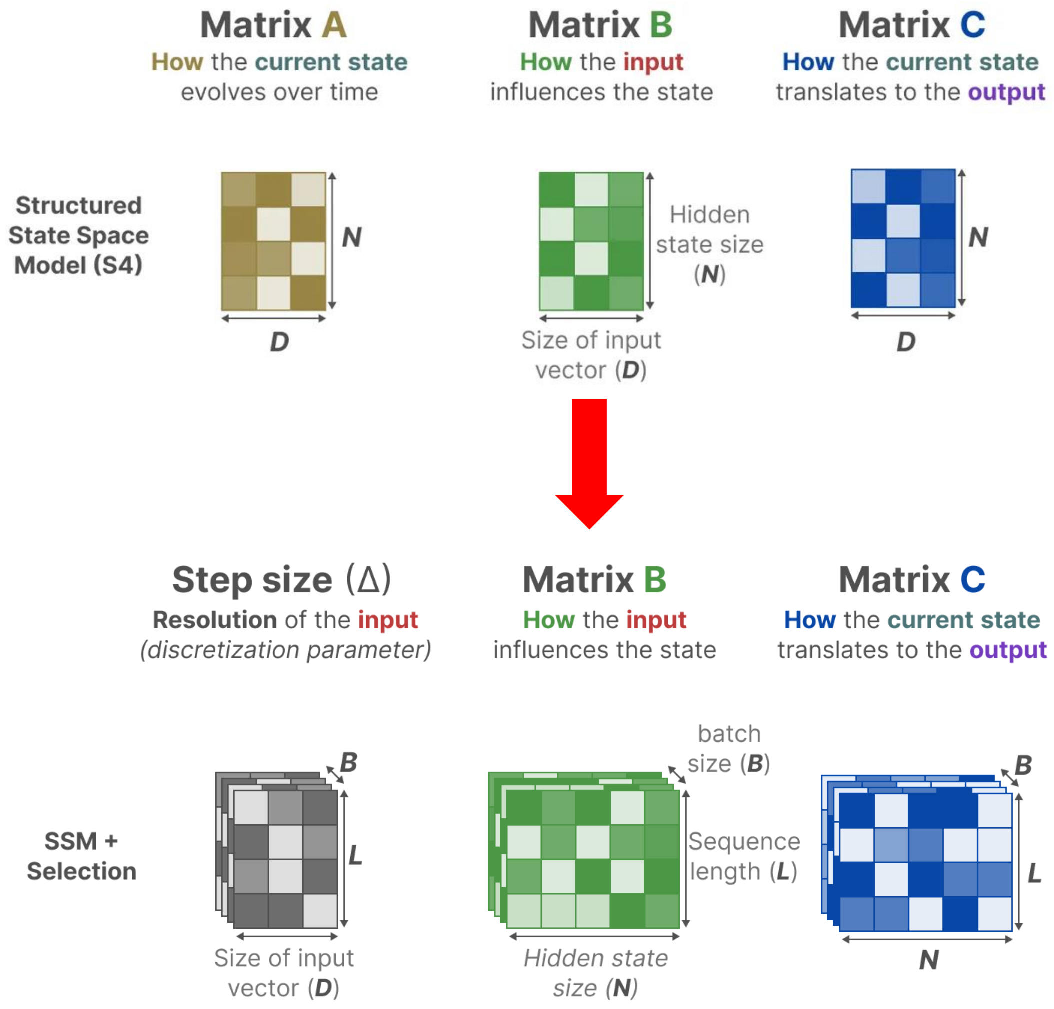
[1] Machine Learning Department, Carnegie Mellon University
[2] Department of Computer Science, Princeton University
agu@cs.cmu.edu, tri@tridao.me

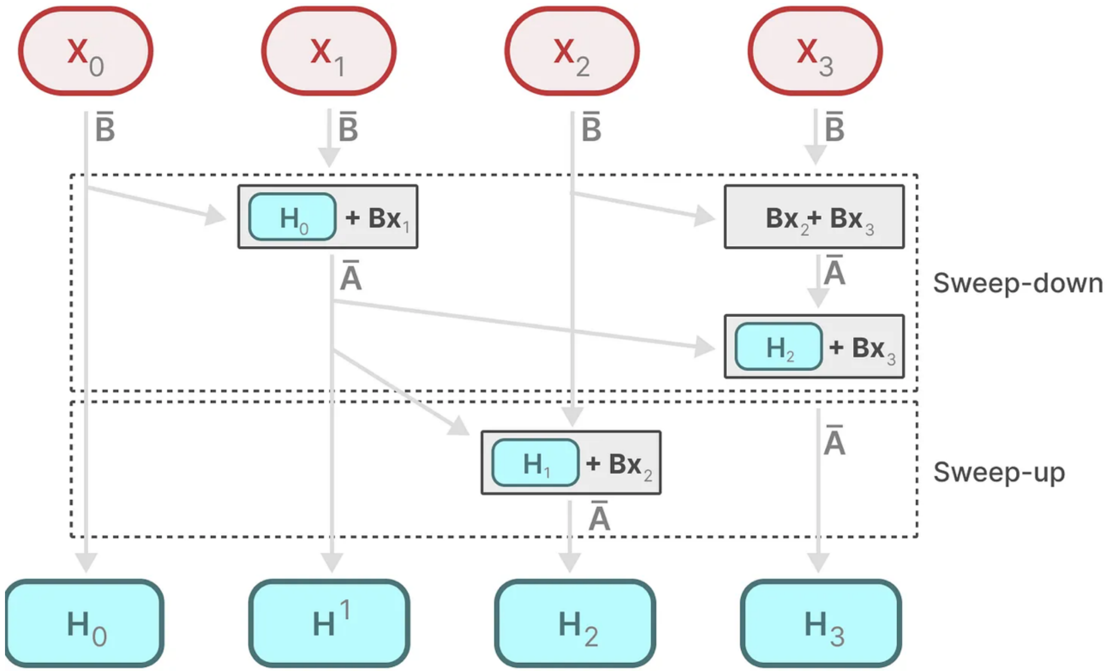
Foundation models, now powering most o [illegible] eep learning, are almost universally based on the Transformer architecture a [illegible] e. Many subquadratic-time architectures such as linear attention, gated convolu [illegible] odels, and structured state space models (SSMs) have been developed to address Trans [illegible] inefficiency on long sequences, but they have not performed as well as attention on in [illegible] language. We identify that a key weakness of such models is their inability to per [illegible] several improvements. First, simply letting the SSM parameters be fun [illegible] t addresse [illegible] vith discrete modalities, allowing the model to selectively propaga [illegible] ormation along the sequence length dimension depending on the current token. Second, eve [illegible] change prevents the use of efficient convolutions, we design a hardware-aware parallel algori [illegible] ent mode. We integrate these selective SSMs into a simplified end-to-end neural network arc [illegible] out attention or even MLP blocks (Mamba). Mamba enjoys fast inference (5× higher throughp [illegible] formers) and linear scaling in sequence length, and its performance [illegible]

Introduction



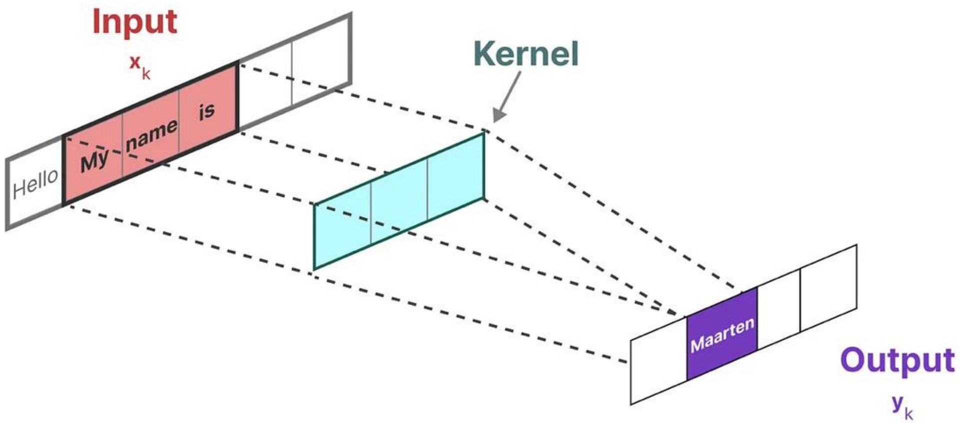
Introduction

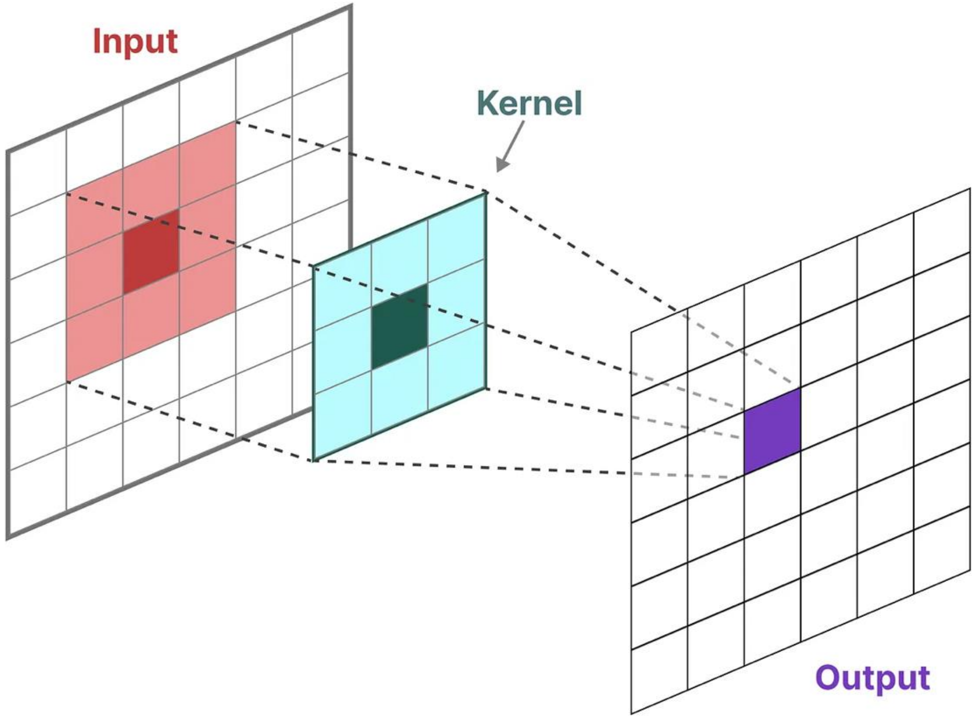




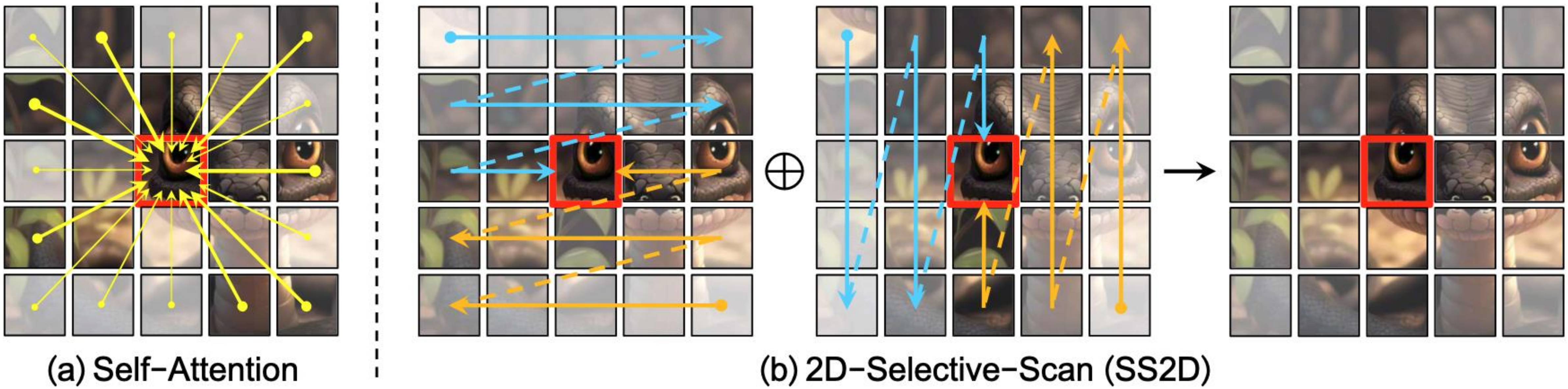
Parallel computation **O(n/t)**

Introduction





2D-Selective-Scan(2DSS)



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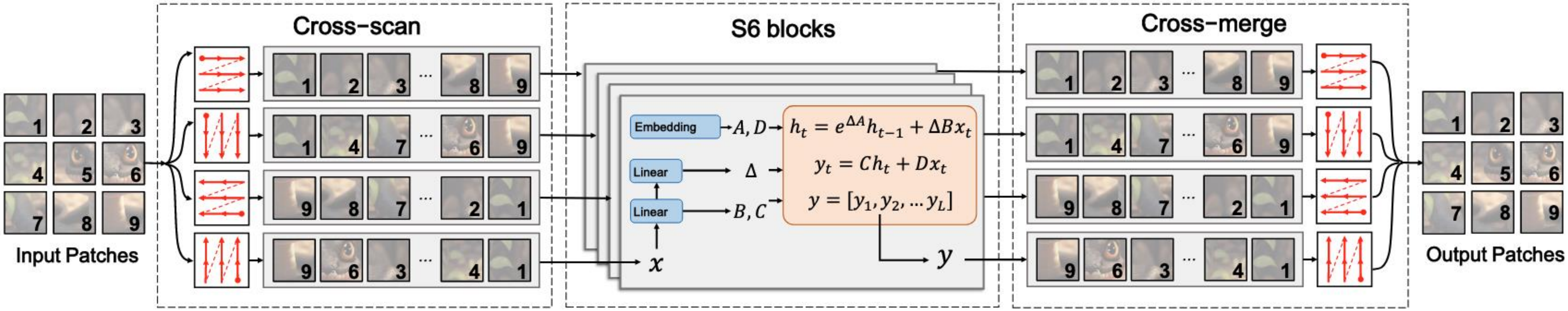
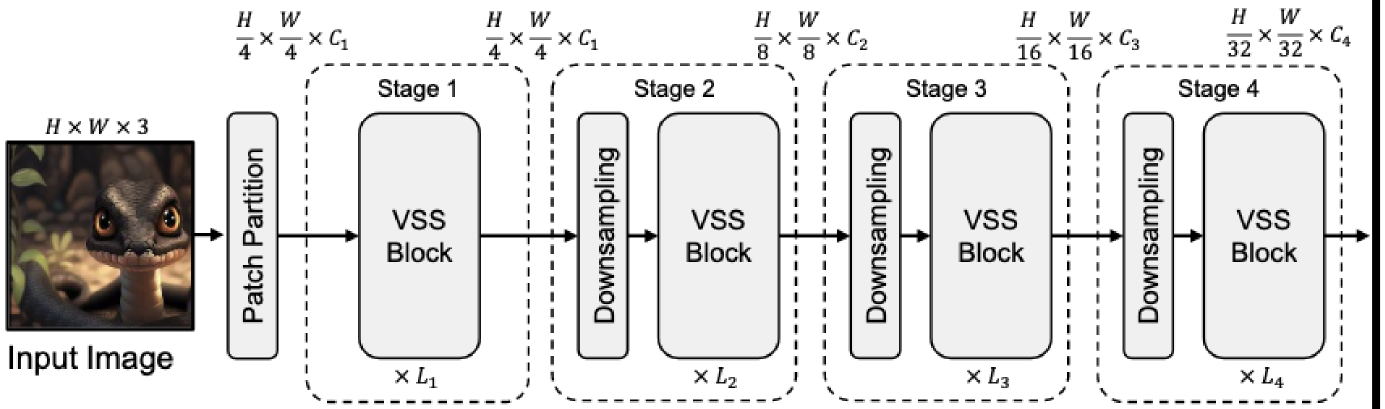
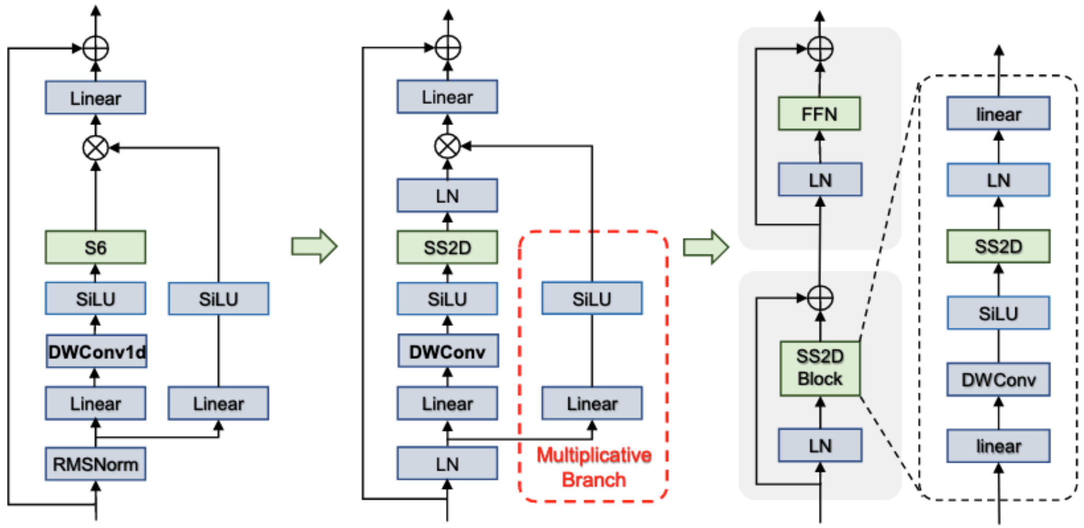
2D-Selective-Scan(2DSS)



Architecture



(a) Architecture of VMamba



(b) Mamba Block (c) The Vanilla VSS Block (d) VSS Block

Architecture

layer name	output size	Vanilla-VMamba-T	Vanilla-VMamba-S	Vanilla-VMamba-B
stem	112×112	conv 4×4, 96, stride 4	conv 4×4, 96, stride 4	conv 4×4, 128, stride 4
stage 1	56×56	vanilla VSSBLock [Linear 96 → 2×96 DWConv 3×3, 2×96 SS2D, dim 2×96 Linear 2×96 → 96 Multiplicative Linear 2×96 → 96] ×2 patch merging → 192	vanilla VSSBLock [Linear 96 → 2×96 DWConv 3×3, 2×96 SS2D, dim 2×96 Linear 2×96 → 96 Multiplicative Linear 2×96 → 96] ×2 patch merging → 192	vanilla VSSBLock [Linear 128 → 2×128 DWConv 3×3, 2×128 SS2D, dim 2×128 Linear 2×128 → 128 Multiplicative Linear 2×128 → 128] ×2 patch merging → 256
stage 2	28×28	vanilla VSSBLock [Linear 192 → 2×192 DWConv 3×3, 2×192 SS2D, dim 2×192 Linear 2×192 → 192 Multiplicative Linear 2×192 → 192] ×2 patch merging → 384	vanilla VSSBLock [Linear 192 → 2×192 DWConv 3×3, 2×192 SS2D, dim 2×192 Linear 2×192 → 192 Multiplicative Linear 2×192 → 192] ×2 patch merging → 384	vanilla VSSBLock [Linear 256 → 2×256 DWConv 3×3, 2×256 SS2D, dim 2×256 Linear 2×256 → 256 Multiplicative Linear 2×256 → 256] ×2 patch merging → 512
stage 3	14×14	vanilla VSSBLock [Linear 384 → 2×384 DWConv 3×3, 2×384 SS2D, dim 2×384 Linear 2×384 → 384 Multiplicative Linear 2×384 → 384] ×9 patch merging → 768	vanilla VSSBLock [Linear 384 → 2×384 DWConv 3×3, 2×384 SS2D, dim 2×384 Linear 2×384 → 384 Multiplicative Linear 2×384 → 384] ×27 patch merging → 768	vanilla VSSBLock [Linear 512 → 2×512 DWConv 3×3, 2×512 SS2D, dim 2×512 Linear 2×512 → 512 Multiplicative Linear 2×512 → 512] ×27 patch merging → 1024
stage 4	7×7	vanilla VSSBLock [Linear 768 → 2×768 DWConv 3×3, 2×768 SS2D, dim 2×768 Linear 2×768 → 768 Multiplicative Linear 2×768 → 768] ×2	vanilla VSSBLock [Linear 768 → 2×768 DWConv 3×3, 2×768 SS2D, dim 2×768 Linear 2×768 → 768 Multiplicative Linear 2×768 → 768] ×2	vanilla VSSBLock [Linear 1024 → 2×1024 DWConv 3×3, 2×1024 SS2D, dim 2×1024 Linear 2×1024 → 1024 Multiplicative Linear 2×1024 → 1024] ×2
	1×1	average pool, 1000-d fc, softmax		
Param. (M)		22.9	44.4	76.3
FLOPs		5.63×10^9	11.23×10^9	18.02×10^9

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Architecture

layer name	output size	VMamba-T	VMamba-S	VMamba-B
stem	112×112	conv 3×3 stride 2, LayerNorm, GeLU, conv 3×3 stride 2, LayerNorm		
stage 1	56×56	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 96 → ssm-ratio ×96 DWConv 3×3, ssm-ratio ×96 SS2D, dim ssm-ratio ×96 Linear ssm-ratio ×96 → 96 FFN mlp-ratio ×96] ×2	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 96 → ssm-ratio ×96 DWConv 3×3, ssm-ratio ×96 SS2D, dim ssm-ratio ×96 Linear ssm-ratio ×96 → 96 FFN mlp-ratio ×96] ×2	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 128 → ssm-ratio ×128 DWConv 3×3, ssm-ratio ×128 SS2D, dim ssm-ratio ×128 Linear ssm-ratio ×128 → 128 FFN mlp-ratio ×128] ×2
		conv 3×3 stride 2, LayerNorm		
stage 2	28×28	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 192 → ssm-ratio ×192 DWConv 3×3, ssm-ratio ×192 SS2D, dim ssm-ratio ×192 Linear ssm-ratio ×192 → 192 FFN mlp-ratio ×192] ×2	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 192 → ssm-ratio ×192 DWConv 3×3, ssm-ratio ×192 SS2D, dim ssm-ratio ×192 Linear ssm-ratio ×192 → 192 FFN mlp-ratio ×192] ×2	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 256 → ssm-ratio ×256 DWConv 3×3, ssm-ratio ×256 SS2D, dim ssm-ratio ×256 Linear ssm-ratio ×256 → 256 FFN mlp-ratio ×256] ×2
		conv 3×3 stride 2, LayerNorm		
stage 3	14×14	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 384 → ssm-ratio ×384 DWConv 3×3, ssm-ratio ×384 SS2D, dim ssm-ratio ×384 Linear ssm-ratio ×384 → 384 FFN mlp-ratio ×384] ×8	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 384 → ssm-ratio ×384 DWConv 3×3, ssm-ratio ×384 SS2D, dim ssm-ratio ×384 Linear ssm-ratio ×384 → 384 FFN mlp-ratio ×384] ×15	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 512 → ssm-ratio ×512 DWConv 3×3, ssm-ratio ×512 SS2D, dim ssm-ratio ×512 Linear ssm-ratio ×512 → 512 FFN mlp-ratio ×512] ×15
		conv 3×3 stride 2, LayerNorm		
stage 4	7×7	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 768 → ssm-ratio ×768 DWConv 3×3, ssm-ratio ×768 SS2D, dim ssm-ratio ×768 Linear ssm-ratio ×768 → 768 FFN mlp-ratio ×768] ×2	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 768 → ssm-ratio ×768 DWConv 3×3, ssm-ratio ×768 SS2D, dim ssm-ratio ×768 Linear ssm-ratio ×768 → 768 FFN mlp-ratio ×768] ×2	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 1024 → ssm-ratio ×1024 DWConv 3×3, ssm-ratio ×1024 SS2D, dim ssm-ratio ×1024 Linear ssm-ratio ×1024 → 1024 FFN mlp-ratio ×1024] ×2
	1×1	average pool, 1000-d fc, softmax		
Param. (M)		30.2	50.1	88.6
FLOPs		4.91×10^9	8.72×10^9	15.36×10^9

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Architectur

layer name	output size	Vanilla-VMamba-T	Vanilla-VMamba-S	Vanilla-VMamba-B
stem	112×112	conv 4×4, 96, stride 4	conv 4×4, 96, stride 4	conv 4×4, 128, stride 4
stage 1	56×56	vanilla VSSBLock [Linear 96 → 2×96 DWConv 3×3, 2×96 SS2D, dim 2×96 Linear 2×96 → 96 Multiplicative Linear 2×96 → 96] ×2	vanilla VSSBLock [Linear 96 → 2×96 DWConv 3×3, 2×96 SS2D, dim 2×96 Linear 2×96 → 96 Multiplicative Linear 2×96 → 96] ×2	vanilla VSSBLock [Linear 128 → 2×128 DWConv 3×3, 2×128 SS2D, dim 2×128 Linear 2×128 → 128 Multiplicative Linear 2×128 → 128] ×2
		patch merging → 192	patch merging → 192	patch merging → 256
stage 2	28×28	vanilla VSSBLock [Linear 192 → 2×192 DWConv 3×3, 2×192 SS2D, dim 2×192 Linear 2×192 → 192 Multiplicative Linear 2×192 → 192] ×2	vanilla VSSBLock [Linear 192 → 2×192 DWConv 3×3, 2×192 SS2D, dim 2×192 Linear 2×192 → 192 Multiplicative Linear 2×192 → 192] ×2	vanilla VSSBLock [Linear 256 → 2×256 DWConv 3×3, 2×256 SS2D, dim 2×256 Linear 2×256 → 256 Multiplicative Linear 2×256 → 256] ×2
		patch merging → 384	patch merging → 384	patch merging → 512
stage 3	14×14	vanilla VSSBLock [Linear 384 → 2×384 DWConv 3×3, 2×384 SS2D, dim 2×384 Linear 2×384 → 384 Multiplicative Linear 2×384 → 384] ×9	vanilla VSSBLock [Linear 384 → 2×384 DWConv 3×3, 2×384 SS2D, dim 2×384 Linear 2×384 → 384 Multiplicative Linear 2×384 → 384] ×27	vanilla VSSBLock [Linear 512 → 2×512 DWConv 3×3, 2×512 SS2D, dim 2×512 Linear 2×512 → 512 Multiplicative Linear 2×512 → 512] ×27
		patch merging → 768	patch merging → 768	patch merging → 1024
stage 4	7×7	vanilla VSSBLock [Linear 768 → 2×768 DWConv 3×3, 2×768 SS2D, dim 2×768 Linear 2×768 → 768 Multiplicative Linear 2×768 → 768] ×2	vanilla VSSBLock [Linear 768 → 2×768 DWConv 3×3, 2×768 SS2D, dim 2×768 Linear 2×768 → 768 Multiplicative Linear 2×768 → 768] ×2	vanilla VSSBLock [Linear 1024 → 2×1024 DWConv 3×3, 2×1024 SS2D, dim 2×1024 Linear 2×1024 → 1024 Multiplicative Linear 2×1024 → 1024] ×2
	1×1	average pool, 1000-d fc, softmax		
Param. (M)		22.9	44.4	76.3
FLOPs		5.63×10^9	11.23×10^9	18.02×10^9

Architecture

layer name	output size	VMamba-T	VMamba-S	VMamba-B
stem	112×112	conv 3×3 stride 2, LayerNorm, GeLU, conv 3×3 stride 2, LayerNorm		
stage 1	56×56	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 96 → ssm-ratio ×96 DWConv 3×3, ssm-ratio ×96 SS2D, dim ssm-ratio ×96 Linear ssm-ratio ×96 → 96 FFN mlp-ratio ×96] ×2	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 96 → ssm-ratio ×96 DWConv 3×3, ssm-ratio ×96 SS2D, dim ssm-ratio ×96 Linear ssm-ratio ×96 → 96 FFN mlp-ratio ×96] ×2	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 128 → ssm-ratio ×128 DWConv 3×3, ssm-ratio ×128 SS2D, dim ssm-ratio ×128 Linear ssm-ratio ×128 → 128 FFN mlp-ratio ×128] ×2
		conv 3×3 stride 2, LayerNorm		
stage 2	28×28	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 192 → ssm-ratio ×192 DWConv 3×3, ssm-ratio ×192 SS2D, dim ssm-ratio ×192 Linear ssm-ratio ×192 → 192 FFN mlp-ratio ×192] ×2	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 192 → ssm-ratio ×192 DWConv 3×3, ssm-ratio ×192 SS2D, dim ssm-ratio ×192 Linear ssm-ratio ×192 → 192 FFN mlp-ratio ×192] ×2	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 256 → ssm-ratio ×256 DWConv 3×3, ssm-ratio ×256 SS2D, dim ssm-ratio ×256 Linear ssm-ratio ×256 → 256 FFN mlp-ratio ×256] ×2
		conv 3×3 stride 2, LayerNorm		
stage 3	14×14	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 384 → ssm-ratio ×384 DWConv 3×3, ssm-ratio ×384 SS2D, dim ssm-ratio ×384 Linear ssm-ratio ×384 → 384 FFN mlp-ratio ×384] ×8	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 384 → ssm-ratio ×384 DWConv 3×3, ssm-ratio ×384 SS2D, dim ssm-ratio ×384 Linear ssm-ratio ×384 → 384 FFN mlp-ratio ×384] ×15	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 512 → ssm-ratio ×512 DWConv 3×3, ssm-ratio ×512 SS2D, dim ssm-ratio ×512 Linear ssm-ratio ×512 → 512 FFN mlp-ratio ×512] ×15
		conv 3×3 stride 2, LayerNorm		
stage 4	7×7	VSSBLock(ssm-ratio=1, mlp-ratio=4) [Linear 768 → ssm-ratio ×768 DWConv 3×3, ssm-ratio ×768 SS2D, dim ssm-ratio ×768 Linear ssm-ratio ×768 → 768 FFN mlp-ratio ×768] ×2	VSSBlock(ssm-ratio=2, mlp-ratio=4) [Linear 768 → ssm-ratio ×768 DWConv 3×3, ssm-ratio ×768 SS2D, dim ssm-ratio ×768 Linear ssm-ratio ×768 → 768 FFN mlp-ratio ×768] ×2	VSSBLock(ssm-ratio=2, mlp-ratio=4) [Linear 1024 → ssm-ratio ×1024 DWConv 3×3, ssm-ratio ×1024 SS2D, dim ssm-ratio ×1024 Linear ssm-ratio ×1024 → 1024 FFN mlp-ratio ×1024] ×2
	1×1	average pool, 1000-d fc, softmax		
Param. (M)		30.2	50.1	88.6
FLOPs		4.91×10^9	8.72×10^9	15.36×10^9

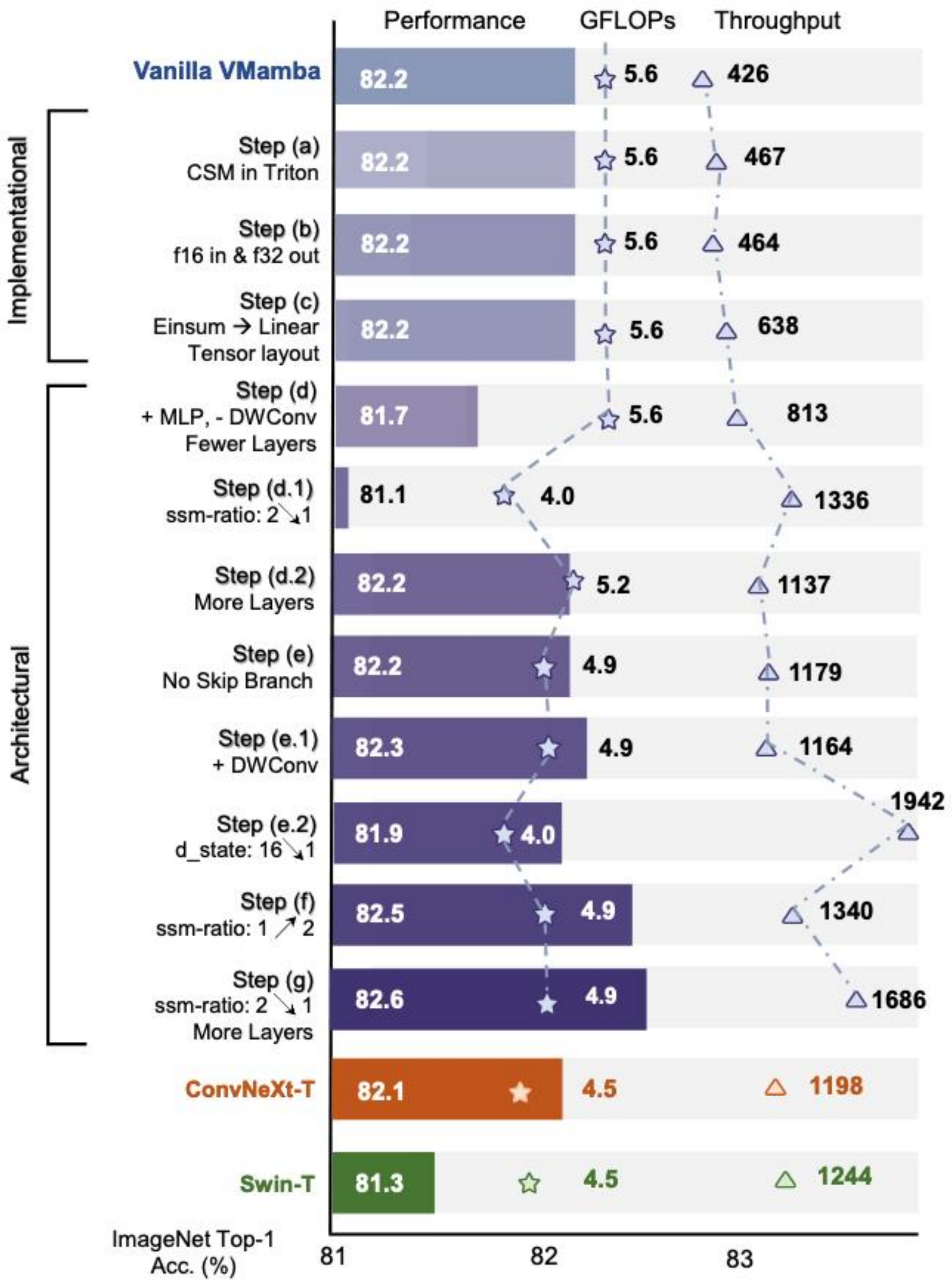
Architecture

Algorithm 2 SSM + Selection (S6)

Input: $x : (\mathsf{B}, \mathsf{L}, \mathsf{D})$
Output: $y : (\mathsf{B}, \mathsf{L}, \mathsf{D})$

1: $\boldsymbol{A} : (\mathsf{D}, \mathsf{N}) \leftarrow \text{Parameter}$
▷ Represents structured $N \times N$ matrix
2: $\boldsymbol{B} : (\mathsf{B}, \mathsf{L}, \mathsf{N}) \leftarrow s_B(x)$
3: $\boldsymbol{C} : (\mathsf{B}, \mathsf{L}, \mathsf{N}) \leftarrow s_C(x)$
4: $\Delta : (\mathsf{B}, \mathsf{L}, \mathsf{D}) \leftarrow \tau_\Delta(\text{Parameter}+s_\Delta(x))$
5: $\overline{\boldsymbol{A}}, \overline{\boldsymbol{B}} : (\mathsf{B}, \mathsf{L}, \mathsf{D}, \mathsf{N}) \leftarrow \text{discretize}(\Delta, \boldsymbol{A}, \boldsymbol{B})$
6: $y \leftarrow \text{SSM}(\overline{\boldsymbol{A}}, \overline{\boldsymbol{B}}, \boldsymbol{C})(x)$
▷ Time-varying: recurrence (*scan*) only
7: **return** y

Experiments



(e) Performance Comparison

Experiments

Model	Params (M)	FLOPs (G)	TP. (img/s)	Top-1 (%)
Transformer-Based				
DeiT-S [57]	22M	4.6G	1761	79.8
DeiT-B [57]	86M	17.5G	503	81.8
HiViT-T [66]	19M	4.6G	1393	82.1
HiViT-S [66]	38M	9.1G	712	83.5
HiViT-B [66]	66M	15.9G	456	83.8
Swin-T [36]	28M	4.5G	1244	81.3
Swin-S [36]	50M	8.7G	718	83.0
Swin-B [36]	88M	15.4G	458	83.5
XCiT-S24 [1]	48M	9.2G	671	82.6
XCiT-M24 [1]	84M	16.2G	423	82.7

Model	Params (M)	FLOPs (G)	TP. (img/s)	Top-1 (%)
ConvNet-Based				
ConvNeXt-T [37]	29M	4.5G	1198	82.1
ConvNeXt-S [37]	50M	8.7G	684	83.1
ConvNeXt-B [37]	89M	15.4G	436	83.8
SSM-Based				
S4ND-Conv-T [44]	30M	5.2G	683	82.2
S4ND-ViT-B [44]	89M	17.1G	397	80.4
Vim-S [69]	26M	5.3G	811	80.5
VMamba-T	30M	4.9G	1686	82.6
VMamba-S	50M	8.7G	877	83.6
VMamba-B	89M	15.4G	646	83.9

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Experiments

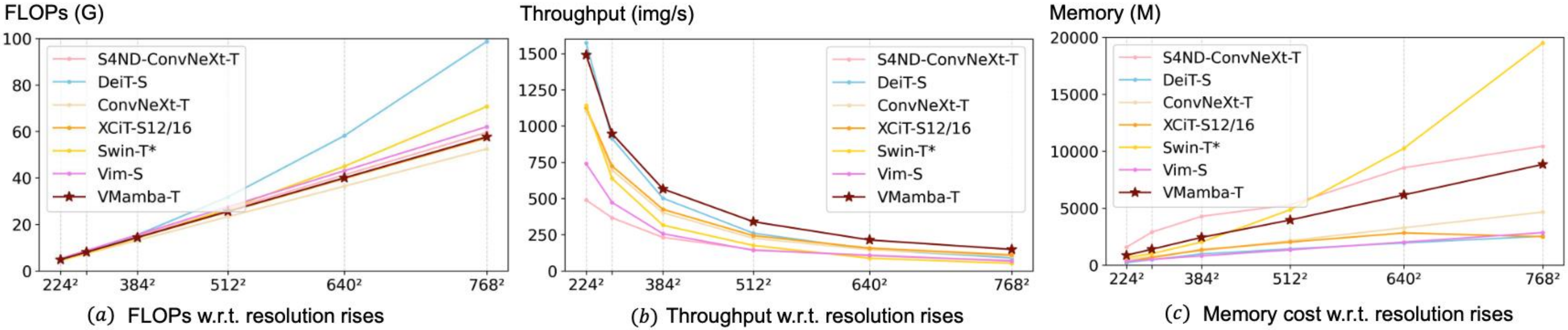
Mask R-CNN 1× schedule				
Backbone	AP^b	AP^m	Params	FLOPs
Swin-T	42.7	39.3	48M	267G
ConvNeXt-T	44.2	40.1	48M	262G
VMamba-T	47.3	42.7	50M	271G
Swin-S	44.8	40.9	69M	354G
ConvNeXt-S	45.4	41.8	70M	348G
VMamba-S	48.7	43.7	70M	349G
Swin-B	46.9	42.3	107M	496G
ConvNeXt-B	47.0	42.7	108M	486G
VMamba-B	49.2	44.1	108M	485G
Mask R-CNN 3× MS schedule				
Swin-T	46.0	41.6	48M	267G
ConvNeXt-T	46.2	41.7	48M	262G
NAT-T	47.7	42.6	48M	258G
VMamba-T	48.8	43.7	50M	271G
Swin-S	48.2	43.2	69M	354G
ConvNeXt-S	47.9	42.9	70M	348G
NAT-S	48.4	43.2	70M	330G
VMamba-S	49.9	44.2	70M	349G

ADE20K with crop size 512				
Backbone	mIOU (SS)	mIOU (MS)	Params	FLOPs
ResNet-50	42.1	42.8	67M	953G
DeiT-S + MLN	43.8	45.1	58M	1217G
Swin-T	44.5	45.8	60M	945G
ConvNeXt-T	46.0	46.7	60M	939G
NAT-T	47.1	48.4	58M	934G
Vim-S	44.9	-	46M	-
VMamba-T	47.9	48.8	62M	949G
ResNet-101	43.8	44.9	86M	1030G
DeiT-B + MLN	45.5	47.2	144M	2007G
Swin-S	47.6	49.5	81M	1039G
ConvNeXt-S	48.7	49.6	82M	1027G
NAT-S	48.0	49.5	82M	1010G
VMamba-S	50.6	51.2	82M	1028G
Swin-B	48.1	49.7	121M	1188G
ConvNeXt-B	49.1	49.9	122M	1170G
NAT-B	48.5	49.7	123M	1137G
RepLKNet-31B	49.9	50.6	112M	1170G
VMamba-B	51.0	51.6	122M	1170G

Experiments

	CoAtNet [4]	A^3 Net [5]	ImageGCN [6]	VMamba−T	VMamba−S	VMamba−B
Average (%)	**84.2**	82.6	83.2	80.4	80.6	82.9

Experiments



Add Unnecessary Details

- Relationship between SS2D and Self-attention

$$\mathbf{V} := [\mathbf{V_1}; \dots; \mathbf{V_T}] \in \mathbb{R}^{T \times D_v}, \text{ where } \mathbf{V_i} := \mathbf{u_{a+i-1}} \odot \boldsymbol{\Delta}_{a+i-1} \in \mathbb{R}^{1 \times D_v}$$
$$\mathbf{K} := [\mathbf{K_1}; \dots; \mathbf{K_T}] \in \mathbb{R}^{T \times D_k}, \text{ where } \mathbf{K_i} := \mathbf{B_{a+i-1}} \in \mathbb{R}^{1 \times D_k}$$
$$\mathbf{Q} := [\mathbf{Q_1}; \dots; \mathbf{Q_T}] \in \mathbb{R}^{T \times D_k}, \text{ where } \mathbf{Q_i} := \mathbf{C_{a+i-1}} \in \mathbb{R}^{1 \times D_k}$$
$$\mathbf{w} := [\mathbf{w_1}; \dots; \mathbf{w_T}] \in \mathbb{R}^{T \times D_k \times D_v}, \text{ where } \mathbf{w_i} := \prod_{j=1}^{i} e^{\mathbf{A}\boldsymbol{\Delta}_{a-1+j}^{\top}} \in \mathbb{R}^{D_k \times D_v}$$
$$\mathbf{H} := [\mathbf{h_{a+1}}; \dots; \mathbf{h_b}] \in \mathbb{R}^{T \times D_k \times D_v}, \text{ where } \mathbf{h_i} \in \mathbb{R}^{D_k \times D_v}$$
$$\mathbf{Y} := [\mathbf{y_{a+1}}; \dots; \mathbf{y_b}] \in \mathbb{R}^{T \times D_v}, \text{ where } \mathbf{y_i} \in \mathbb{R}^{D_v}$$

$$\mathbf{h_b} = \mathbf{w_T} \odot \mathbf{h_a} + \sum_{i=1}^{T} \frac{\mathbf{w_T}}{\mathbf{w_i}} \odot \left(\mathbf{K_i}^{\top} \mathbf{V_i}\right),$$
$$\mathbf{y_b} = \mathbf{Q_T} \mathbf{h_b}$$
$$= \mathbf{Q_T} \left(\mathbf{w_T} \odot \mathbf{h_a}\right) + \mathbf{Q_T} \sum_{i=1}^{T} \frac{\mathbf{w_T}}{\mathbf{w_i}} \odot \left(\mathbf{K_i}^{\top} \mathbf{V_i}\right)$$

$$\mathbf{y_b}^{(j)} = \left(\mathbf{Q_T} \odot \mathbf{w_T}^{(j)}\right) \mathbf{h_a}^{(j)} + \sum_{i=1}^{T} \left(\frac{\mathbf{Q_T} \odot \mathbf{w_T}^{(j)}}{\mathbf{w_i}^{(j)}} \mathbf{K_i}^{\top}\right) \odot \mathbf{V_i}^{(j)}$$

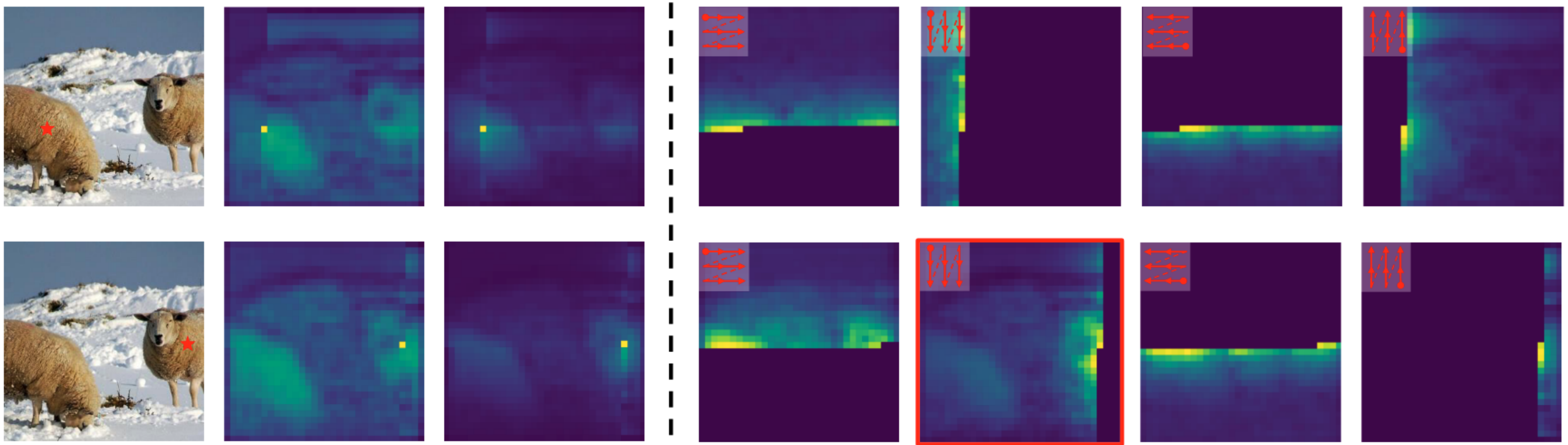
Add Unnecessary Details

- Relationship between SS2D and Self-attention

$$\mathbf{Y}^{(j)} = \left(\mathbf{Q} \odot \mathbf{w}^{(j)}\right) \mathbf{h_a}^{(j)} + \left[\left(\mathbf{Q} \odot \mathbf{w}^{(j)}\right) \left(\frac{\mathbf{K}}{\mathbf{w}^{(j)}}\right)^{\top} \odot \mathbf{M}\right] \mathbf{V}^{(j)}$$

Add Unnecessary Details

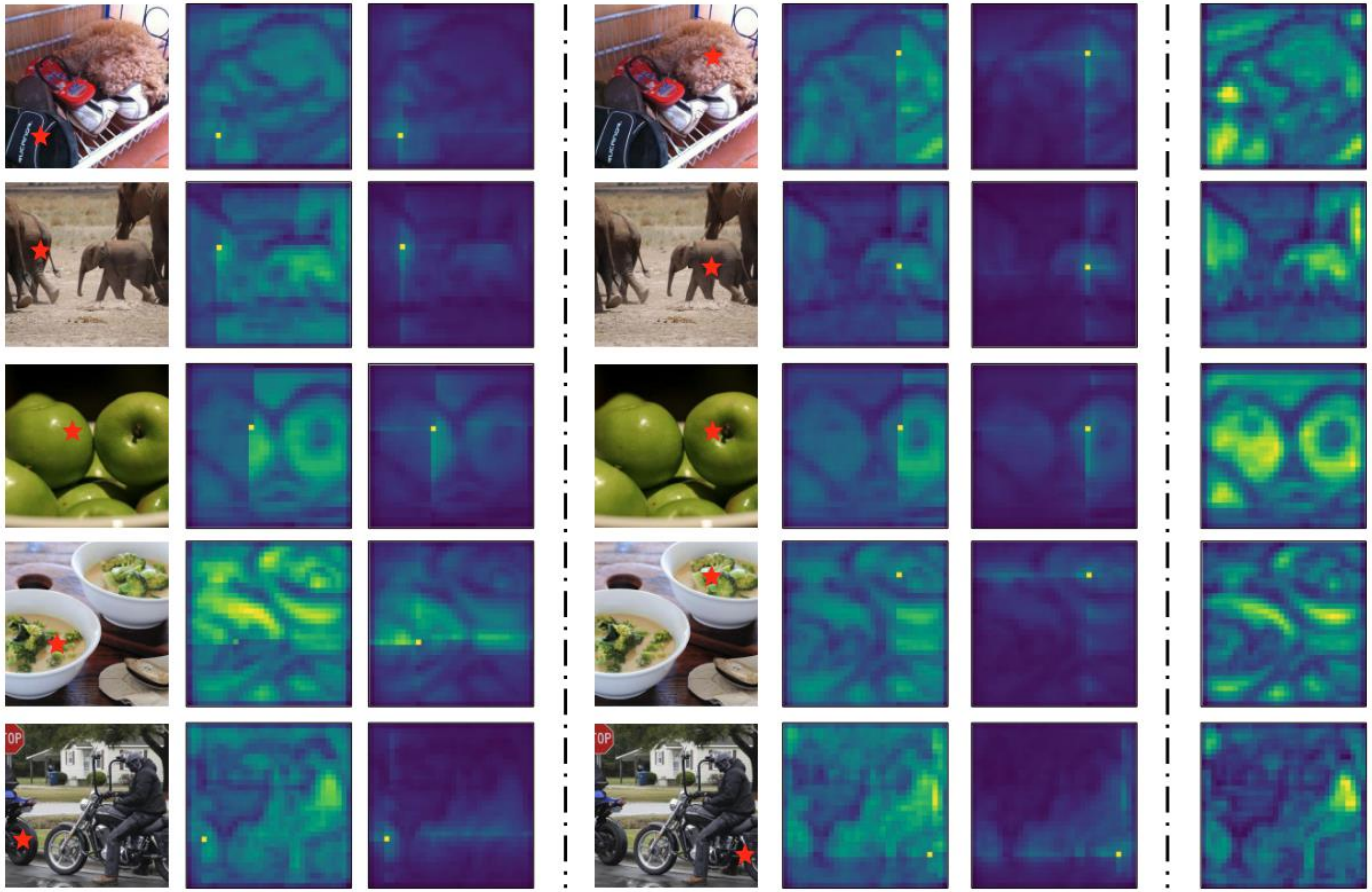
- Visualization of Attention and Activation Maps



(a) Input Image (b) Activation map of QK^T (c) Activation map of $(Q \odot w)(K/w)^T$ (d) Activation map of $(Q \odot w)(K/w)^T$ for individual scanning path

Add Unnecessary Details

- Visualization of Attention and Activation Maps



(a) Input Image (b) Activation Map of QK^T (c) Activation Map of $(Q \odot w)(\frac{K}{w})^T$ (d) Input Image (e) Activation Map of QK^T (f) Activation Map of $(Q \odot w)(\frac{K}{w})^T$ (g) Diagonal of Attention Map

HAI
Human-centered AI


Thank You!